

04. Browser Database Schema: PrismDB (Figure 4.4)

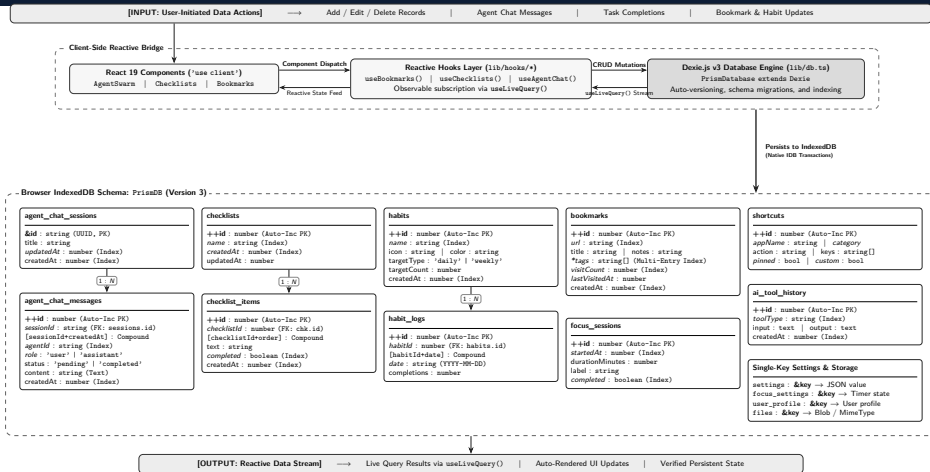


Figure 4.4: Browser IndexedDB Database Schema and Reactive Data Architecture (PrismDB)

Database Schema & Reactive Engine Glossary

PrismDB: Local client-side DB built on IndexedDB storing sessions, habits, tasks, & settings.
IndexedDB (IDB): W3C standard browser NoSQL store providing transactional persistence.
Dexie.js v3: Minimalist TypeScript wrapper handling migrations, indices, and transactions.

PK (++)id / &id: Primary Key; ++id auto-increments integers; &id unique string/UUID.
idx / Compound Index: Secondary B-tree index; [a+b] enables high-speed compound filtering.
Multi-Entry (*tags): Array index where each element is indexed individually for membership search.

useLiveQuery(): Reactive hook subscribing React UI to live queries with auto-rendering.
FK (Foreign Key): Relational link associating child entry records with parent table (1 : N).
CRUD Transactions: Atomic Create, Read, Update, Delete executed with ACID guarantees in-browser.